Engineering Guide to Thread Design & CAD Modeling

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Contents

1 Introduction

The first step is always to understand what product you are building for and how your product will be used. If you understand the necessities and constraints of the product, you can make thread-design decisions much faster and avoid unnecessary redesign later. Before selecting a thread, consider factors like expected loads, materials being joined, available space, required strength, operating environment, assembly and disassembly frequency, manufacturing method, and whether the connection needs to provide sealing or resist vibration. Once these requirements are understood, you can determine the appropriate thread standard, diameter, pitch, tolerance class, engagement length, and other design parameters.

2 Define Thread Specs and Full Thread Callout

A thread callout is a standardized notation for defining characteristics on a thread. There are two primary types of thread callouts: Metric and Imperial (the American inch-based system).

2.1 Metric Callout

Example: `M12 x 1.75 x 85 6g Screw`

- **M**: Designates a Metric Thread.
- **12**: Major Diameter in millimeters (12 mm). This is the diameter to the outermost crests of the threads.
- **1.75**: Thread Pitch in millimeters (1.75 mm). Pitch represents the distance between the peaks of two adjacent threads.
- **85**: Fastener Length in millimeters (85 mm). This value is sometimes omitted in simple callouts.
- **6g:** Tolerance class. The lowercase letter tells us this is an external thread with medium grade tolerance. More on tolerance in following sections

2.2 Imperial Callout

Example: `#4-40 UNC-3A x 0.5 Screw`

- **#4**: Major Diameter size designation. To find the exact numerical diameter, refer to standard size charts. If no '#' prefix is present (e.g., `2-40 UNC-3A`), the first number directly specifies the major diameter in inches (e.g., 2 inches).
- **40**: Threads Per Inch (TPI). In this case, there are 40 threads per inch along the shaft. The relationship between pitch (P) and TPI is:

$$P = \frac{1}{\text{TPI}}$$

- **UNC**: Thread Standard designation (Coarse series).
  I. **UNC (Coarse):** Best for general-purpose applications, softer materials, faster assembly, and lower TPI counts.
  II. **UNF (Fine):** Higher TPI, suitable for thin wall applications, precise clamp-load control, and elevated vibration resistance.
  III. **UNEF (Extra Fine):** Niche applications, rarely used in general hardware.

IV. *Rule of Thumb:* Use fine threads for thick gauge and robust metal applications; use coarse threads for thin gauge metals and brittle materials (e.g., plywood, OSB, SPF).

- **3A**: Tolerance class designation. The number('3' in this case) describes the fit grade. The range is from 1-3; '1' being a loose fit and '3' being a high precision one. The letter describes the thread type; 'A' is designated for external threads while 'B' is designated for internal threads.
- **RH/LH**: Right-Handed/Left-Handed thread indicator (usually omitted). Right-handed threads tighten clockwise and loosen counter-clockwise(conventional: righty-tighty, lefty-loosey). There are also left-handed threads which tighten counter-clockwise and loosen clockwise.
- **0.5**: Screw shaft length in inches (0.5 inches).

2.3 Introduction to Tolerances

Although full tolerance specs are often not included in basic callouts, understanding them is extremely crucial when modeling custom threads in CAD programs (like SOLIDWORKS).

Because physical manufacturing cannot achieve perfect nominal dimensions, tolerance defines the boundary of deviation within which a part is still functional. A tolerance class consists of a number and a letter:

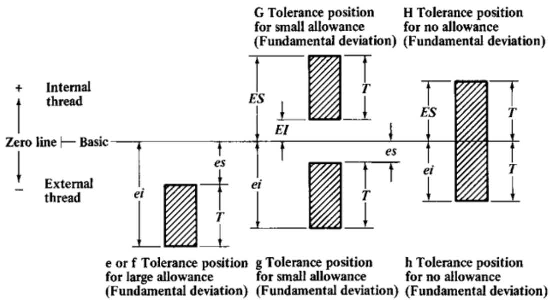

Figure 1: Metric Tolerance System for Screw Threads

- **Tolerance Grade (Number):** A smaller number indicates a tighter tolerance; a larger number indicates a looser tolerance.
- **Tolerance Position (Letter):** Represents the fundamental deviation (shift relative to the basic profile):
- **Internal Threads (e.g., Nuts):** Capital letters (H, G, etc.). H represents zero fundamental deviation, while G is shifted positive (above basic size).
- **External Threads (e.g., Bolts):** Lowercase letters (h, g, f, e, d, c, b, a). h represents zero fundamental deviation; g through a represent progressively larger negative deviations.

  Important: Unified thread classes and ISO metric tolerance grades use different conventions. For Unified threads, Class 3 is tighter than Class 2 or Class 1. For ISO metric threads, a lower numerical tolerance grade generally represents a narrower tolerance zone.

3 Understand Thread Geometry and Definitions

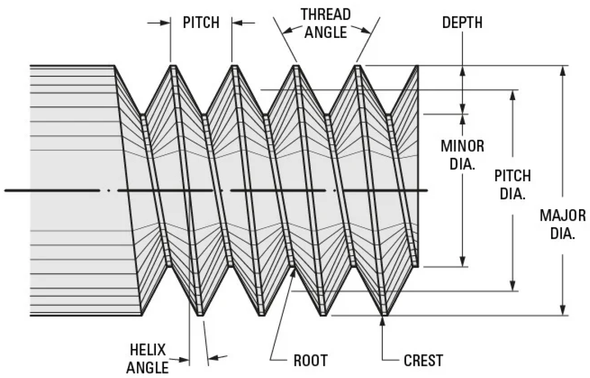

Figure 2: Basic Thread Geometry and Terminology

- **Thread Angle / Included Angle:** The angle formed between adjacent flanks of a thread profile. Standard ISO-Metric and Imperial threads utilize a 60° included angle. Whitworth threads use 55°.
- **Flank Angle:** The angle between one side of a thread flank and a line perpendicular to the thread axis (half of the included angle, typically 30°).
- **Pitch (P):** Axial distance from corresponding points, crest to crest. Governs thread coarseness.
- **Lead (L):** The axial distance a screw advances during one complete revolution.
- **Major (Crest) Diameter:** The largest diameter of an internal or external thread profile.
- **Pitch Diameter (D_2 / d_2):** The diameter of an imaginary coaxial cylinder where the width of the thread ridge equals the width of the thread groove.
- **Root (Minor) Diameter:** The smallest diameter of a thread profile (measured at the root valley).
- **Root Radius (R):** The radius at the root groove. Allows for decreased stress concentrations and standard tool radii.
- **Thread Height:** The radial depth measured from thread root valley to crest.
- **Thread Engagement Length:** The distance over which the internal/external threads are in contact with one another.
- **Male vs. Female Threads:** External threads (bolts/screws) feature outward projections and are designated male. Internal threads (nuts/tapped holes) feature cuts into the material wall and are designated female.
- **Handedness:** Indicates the directional wrap of the threads around the shaft axis (Right-hand standard vs. Left-hand reverse).
- **Parallel vs. Tapered Threads:** Parallel threads maintain constant major/minor diameters along their length and require secondary seals (gaskets/O-rings) for fluid tight applications. Tapered threads gradually reduce in diameter along their length, creating a mechanical wedge/seal (common in pipe fittings).

4 Determine Tolerance Classes and Fundamental Deviations

Select tolerance classes based on functional performance followed by manufacturing feasibility. Common class examples include `6H`, `6g`, and `4g6g`.

When dual callouts like `4g6g` are present for external threads:

- First pair (`4g`) = Pitch diameter tolerance grade and position.
- Second pair (`6g`) = Major diameter tolerance grade and position.
- Single callout (`6g`) = Applies to both pitch and major diameters simultaneously.

Standard combinations typically mate internal `H` positions (zero allowance) with external `g` positions (small allowance for ease of assembly). Refer to standard machinery tables (e.g., Machinery's Handbook) for specific allowances (EI, es) and tolerance limits ($T_{D1}, T_{D2}, T_d, T_{d2}$).

5 CAD Thread Modeling

5.1 Key Parameters Required for Modeling

- **Internal Threads:** Min/Max Minor Diameter, Min/Max Pitch Diameter.
- **External Threads:** Min/Max Major Diameter, Min/Max Pitch Diameter, Max Minor Diameter.

Modeling Strategy: When designing custom threads, standard practice uses the Minimum Minor and Minimum Pitch Diameters when modeling internal features, and Maximum Major and Maximum Pitch Diameters when modeling external features, while calling out tolerances directly on technical drawings. This allows us engineers to simply add unilateral tolerances on our thread drawings that either only increase or decrease based on the type of thread.

5.2 Standard Metric Calculation Formulas

The following equations are from the 27th edition of the Machinery's Handbook, please also verify your latest edition or most applicable standard to consider any recent industry changes.
Internal Threads:

$$\begin{aligned}
\text{Min Major Dia.} &= \text{Basic Major Dia.} + EI \\
\text{Min Pitch Dia.} &= \text{Basic Major Dia.} - 0.649519P + EI_{D2} \\
\text{Max Pitch Dia.} &= \text{Min Pitch Dia.} + T_{D2} \\
\text{Max Major Dia.} &= \text{Max Pitch Dia.} + 0.793857P \\
\text{Min Minor Dia.} &= \text{Min Major Dia.} - 1.082532P \\
\text{Max Minor Dia.} &= \text{Min Minor Dia.} + T_{D1}
\end{aligned}$$

External Threads:

$$\begin{aligned}
\text{Max Major Dia.} &= \text{Basic Major Dia.} - es \\
\text{Min Major Dia.} &= \text{Max Major Dia.} - T_d \\
\text{Max Pitch Dia.} &= \text{Basic Major Dia.} - 0.649519P - es_{d2} \\
\text{Min Pitch Dia.} &= \text{Max Pitch Dia.} - T_{d2} \\
\text{Max Flat Form Minor Dia.} &= \text{Max Pitch Dia.} - 0.433013P \\
\text{Min Rounded Root Minor Dia.} &= \text{Min Pitch Dia.} - 0.616025P \\
\text{Min Root Radius} &= 0.125P
\end{aligned}$$

Verification Step: Always cross-reference non-standard thread calculations against standard sizes in reference tables (e.g., checking custom `M46x1.5` against tabular `M45x1.5` to confirm a consistent +1 mm shift across all dimensions).

5.3 Step-by-Step CAD Modeling Workflow

At this point you are ready to start modeling the threads in your CAD software of choice. For the remainder of this document I will speak in terms of SOLIDWORKS usage as it is what is most comfortable for me.

1. Create a construction helix on a cylindrical base using the appropriate pitch diameter (Max Pitch for external threads, Min Pitch for internal threads).

2. Create a profile sketch on a plane normal/perpendicular to the start of the helix path. Sketch the fundamental 60° thread profile, beginning from an equilateral-triangle construction.
3. Point the apex of the profile inward towards the center axis for external thread cut profiles, or outward for internal threads. The orientation of the sketch profile results in the direction of cut that we will produce.
4. Add the required root radius arc.
5. Add a vertical centerline through the center of the triangle connecting top and bottom sides, assigning it a construction length of $P/2$.
6. Fully constrain the profile by applying the calculated Min Minor / Max Major values or distance to pitch axis.
7. Apply a Pierce constraint between the sketch profile reference point and the helix line.
8. Perform a **Swept Cut** feature along the helix path.
9. If creating a thread pair: Verify that your threads when assembled, have standard clearances between them and are NOT interfering. You can see this in an assembly. Please note however that CAD is still a *simulation* and real world cases may be different. Triple check calculations.

5.4 Special Engineering Cases

- **Plating and Coating:** Applied surface finishes reduce thread clearances. When parts are to be plated, you may need to expand the initial fundamental deviations (e.g., switch from `6H/6g` to `6G/6e` or `6f`). Verify that clearance accounts for approximately 4× the coating thickness on the pitch diameter.
- **High-Temperature Applications:** Differing thermal expansion coefficients or rapid thermal gradients cause binding and galling. Utilize looser classes with larger allowances (e.g., `6G` internal and `6e` external) to maintain adequate clearance under elevated thermal expansion.

6 Manufacturing and Assembly Considerations

- **Tooling Capabilities:** Confirm whether specified tolerances match manufacturing equipment limitations (e.g., lathes, taps, dies, CNC thread milling). Overly tight tolerances increase scrap and cost. Philip will not like this.
- **Machinable Root Radii:** Cutting tools feature natural tip radii. Avoid specifying internal root radii smaller than standard cutting tool radius parameters.
- **Tool Access and Clearance:** Ensure sufficient clearance for assembly tools (wrenches, drivers) and physical hand access within the final assembly environment.
- **Operational Environment:** Evaluate operating forces and environmental hazards (vibration-induced loosening, cyclic fatigue, shear forces, corrosive environments) to select proper thread pitches, locking mechanisms, and materials.

7 BONUS SECTION: O-Ring Selection Principles

7.1 Imperial Callout (AS568 Standard)

Format: `O-Ring AS568-[Dash Number], CS [Cross Section] in x ID [Inner Diameter] in, [Material] [Hardness]`

Table 1: AS568 Dash Number Cross-Section Categories

Dash Series	Nominal Cross-Section (CS)
0xx	0.070 in ($\approx 1/16$ in)
1xx	0.103 in
2xx	0.139 in
3xx	0.210 in
4xx	0.275 in

7.2 Metric Callout

Format: `O-Ring CS [Cross Section] mm x ID [Inner Diameter] mm [Material] [Hardness]`

7.3 General Design Formula & Materials

$$\text{Outer Diameter (OD)} = \text{ID} + (2 \times \text{CS})$$

Common Materials:

- **Buna-N (NBR):** Standard general-purpose material, excellent resistance to petroleum oils and fluids.
- **Viton (FKM):** Outstanding high-temperature performance and broad chemical resistance.
- **EPDM:** Excellent performance with water, steam, coolant, and outdoor/ozone exposure.
- **Silicone (VMQ):** Wide operational temperature window; lower abrasion resistance.